

PreLab 7

At 8 kHz:

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(8k)(10n)} = \boxed{1989.43}$$

$$X_L = 2\pi f L = 2\pi(8k)(10m) = \boxed{502.65}$$

~~$$Z = \sqrt{\left(\frac{1}{R}\right)^2 + \left(\frac{1}{X_L} - \frac{1}{X_C}\right)^2} = \frac{1}{\sqrt{\left(\frac{1}{1000}\right)^2 + \left(\frac{1}{502.65} - \frac{1}{1989.43}\right)^2}} = \boxed{893.47}$$~~

$$B_L = \frac{1}{X_L} = 1.99 \times 10^{-3}, \quad B_C = \frac{1}{X_C} = 5.02 \times 10^{-4}$$

$$Z = \frac{1}{\sqrt{\left(\frac{1}{R}\right)^2 + (B_L - B_C)^2}} = \boxed{557.79 \Omega}$$

~~$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$~~

$$\phi = \cos^{-1} \left(\frac{Z}{R} \right) = \boxed{56.1^\circ}$$

At 12 kHz:

$$X_C = 1326.29, \quad X_L = 753.98$$

$$B_C = 7.54 \times 10^{-4}, \quad B_L = 1.33 \times 10^{-3}$$

$$Z = 866.53 \Omega$$

$$\phi = \cos^{-1} \left(\frac{866.53}{1000} \right) = 29.94^\circ$$

At 16k Hz :

$$X_C = 994.72, \quad X_L = 1005.31$$

$$B_C = 1.005 \times 10^{-3}, \quad B_L = 9.95 \times 10^{-4}$$

$$Z = 999.95 \Omega$$

$$\phi = \cos^{-1} \left(\frac{999.95}{1000} \right) = 0.573^\circ$$

At 20k Hz :

$$X_C = 795.77, \quad X_L = 1256.64$$

$$B_C = 1.23 \times 10^{-3}, \quad B_L = 7.96 \times 10^{-4}$$

$$Z = 917.33 \Omega$$

$$\phi = \cos^{-1} \left(\frac{917.33}{1000} \right) = 23.46^\circ$$

At 24k Hz :

$$X_C = 663.15, \quad X_L = 1507.96$$

$$B_C = 1.51 \times 10^{-3}, \quad B_L = 6.63 \times 10^{-4}$$

$$Z = 763.07, \quad \phi = \cos^{-1} \left(\frac{763.07}{1000} \right) = 40.26^\circ$$

Date: _____

At 30 kHz:

$$X_C = 530.52, \quad X_L = 1884.96$$

$$b_C = 1.88 \times 10^{-3}, \quad B_L = 5.3 \times 10^{-4}$$

$$Z = 595.23 \Omega, \quad \phi = \cos^{-1} \left(\frac{595.23}{1000} \right) = 55.47^\circ$$

Freq	$X_C < \phi$	$X_L < \phi$	$Z < \phi$	ϕ
8K	1989.43	502.65	557.79	56.1°
12K	1326.29	753.98	866.53	29.94°
16K	994.72	1005.31	999.95	0.573°
20K	795.77	1256.64	917.33	23.46°
24K	663.15	1507.96	763.07	40.26°
30K	530.52	1884.96	595.23	55.47°